The Econometrics of Factor Models

PhD Departmental Reading Group

Facilitator: Mr. Daniel Tiong
daniel.tiong@unimelb.edu.au
Department of Economics

Overview and Details

Target Audience. The reading group is aimed at graduate students in economics with a strong frequentist/cross-sectional background (i.e., a background based on the core econometrics sequences undertaken by students within the Department), albeit with relatively less experience in time series or Bayesian econometrics.

Workload. Sessions ranging between 1 to 1.5 hours, 1-2 substantial papers per week, roughly 12 weeks in total.

Structure. The first week aims at the foundations of factor modelling to bring all interested parties up to speed. We then explore classic papers in the factor modelling literature, as well as standard estimation methods. Weeks 5-6 cover Bayesian methods. The remainder of the reading group then begins focusing on empirical challenges encountered in the factor literature, as well as more recent breakthroughs with respect to machine learning methods.

The general guidelines for a session with one paper allocated are: 40-minute presentation, 20-minute discussion.

For two papers, each paper has allocated a 30-minute presentation, 15-minute discussion.

Communication. The timing and venues for weekly sessions are TBA. Each paper will have a main presenter. Additional discussants are also encouraged for each paper of the reading group.

The allocation of papers is determined by a Google sheet; first-come, first-served.

Tentative Reading List

Week 1: Introducing Factor Modelling

References:

- **Stock, J.H. & Watson, M.W. (2002)**. "Macroeconomic Forecasting Using Diffusion Indexes." *Journal of Business & Economic Statistics*, 20(2), 147-162.

Brief overview: Introduces the concept of summarizing massive datasets into a few dynamic components via principal components (PCA). A gentle shift into time-series macro for students used to cross-sectional regressions. A later book chapter by Stock and Watson in 2016 is also considered.

Week 2: Estimation & Identification in Dynamic Factor Models

References:

- **Forni, M., et al. (2000)**. "The Generalized Dynamic-Factor Model: Identification and Estimation." *ReStat*.
- **Bai, J. & Ng, S. (2002)**. "Determining the Number of Factors in Approximate Factor Models." *Econometrica*.

Brief overview: Bai & Ng is the quintessential paper on choosing the "correct" number of factors. Forni et al. introduces the frequency domain vs time domain approach.

Week 3: Classic Asset Pricing Factors

References:

- **Fama, E.F. & French, K.R. (1993)**. "Common risk factors in the returns on stocks and bonds." *JFE*.
- **Fama, E.F. & French, K.R. (2015)**. "A five-factor asset pricing model." *JFE*.

Because of the strongly related scope of these two papers, I am treating both of these papers as a singular one for the purposes of this reading group.

Brief overview: Anchors the topic in traditional empirical asset pricing. Shows how latent statistical factors contrast with theory-motivated, observable economic characteristics.

Week 4: Factor-Augmented Vector Autoregressions (FAVAR)

References:

- **Bernanke, B.S., Boivin, J. & Eliasz, P. (2005)**. "Measuring the Effects of Monetary Policy: A Factor-Augmented Vector Autoregressive (FAVAR) Approach." *QJE*.

Brief overview: Demonstrates how to embed a large information set into structural VARs to properly identify shocks and solve puzzles (like the price puzzle) derived from omitted variables.

Week 5: Bayesian Approaches to Factor Models

References:

- **Kose, M.A., Otrok, C. & Whiteman, C.H. (2003)**. "International Business Cycles: World, Region, and Country-Specific Factors." *The American Economic Review.*

Brief overview: Demystifies Gibbs sampling for estimating unobserved hierarchical factors. Excellent intuition-building for MCMC.

Week 6: High-Dimensional Bayesian VARs vs Factors

References:

- **Bańbura, M., Giannone, D. & Reichlin, L. (2010)**. "Large Bayesian vector auto regressions." *Journal of Applied Econometrics.*

Brief overview: Contrasts shrinking a massive system using Bayesian priors against simply extracting factors, mathematically showing the links between BVARs and factor structures.

Week 7: Weak vs. Strong Factors

References:

- **Chudik, A., Pesaran, M.H. & Tosetti, E. (2011)**. "Weak and strong cross-section dependence and estimation of large panels." *The Econometrics Journal.*
- **Lettau, M. & Pelger, M. (2020)**. "Estimating latent asset-pricing factors." *Journal of Econometrics.*

Brief overview: Differentiates factors affecting all series deeply (strong) versus those with fading impacts (weak). Crucial distinction for asymptotic behavior in panels.

Week 8: Time-Varying Loadings and Structural Breaks

References:

- **Bates, B.J., et al. (2013)**. "Consistent factor estimation in dynamic factor models with structural instability." *Journal of Econometrics.*

Brief overview: Traces what goes wrong (and analytically what goes right) when factor loadings undergo structural changes—very common in post-crisis macro data.

Week 9: Latent vs. Observable Factors & Omitted Variable Bias

References:

- **Giglio, S. & Xiu, D. (2021)**. "Asset Pricing with Omitted Factors." *Journal of Political Economy.*

Brief overview: Bridges latent PCA with observable traits. Innovatively demonstrates how to resolve omitted-variable bias in risk models by extracting an unbiased risk premium for an observable factor.

Week 10: Mixed-Frequency Data and Nowcasting

References:

- **Aruoba, S.B., Diebold, F.X. & Scotti, C. (2009)**. "Real-Time Measurement of Business Conditions." *JBES*.

Brief overview: Shows how dynamic factor models natively accommodate missing data and mixed reporting frequencies via Kalman filtering. Foundational for modern nowcasting.

Week 11: Instrumenting and Taming the 'Factor Zoo'

References:

- **Feng, G., Giglio, S. & Xiu, D. (2020)**. "Taming the Factor Zoo: A Test of New Factors." *The Journal of Finance*.

Supporting reference:

- **Bai, J., and Ng, S. (2019)**. "Rank regularized estimation of approximate factor models." *Journal of Econometrics*.

Brief overview: Addresses the proliferation of discovered factors. Combines machine learning (LASSO) with a factor framework to rigorously test whether a newly proposed factor genuinely offers incremental explanatory value.

Week 12: Factors in Machine Learning

References:

- **Gu, S., Kelly, B. & Xiu, D. (2020)**. "Empirical Asset Pricing via Machine Learning." *RFS*.
- **Pelger, M. & Xiong, R. (2022)**. "State-Varying Factor Models of Large Dimensions." *JBES*.

Brief overview: Examines how deep neural networks and machine learning generalize the traditional linear factor approach, helping models to capture non-linear, dynamic loadings seamlessly.